

A two-dimensional operator of minimal isometry order three

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Conjecture 00000008576

Both source versions assert that the minimal order m of an m -isometric operator is bounded above by the dimension of its space. We disprove this clause with a genuine operator on the complex Euclidean Hilbert space $H = \mathbb{C}^2$.

Definitions and the actual operator

The inner product and norm on H are

$$\langle v, w \rangle = \bar{v}_0 w_0 + \bar{v}_1 w_1, \quad \|v\|^2 = |v_0|^2 + |v_1|^2.$$

The standard basis is orthonormal. For a bounded linear operator T , define

$$B_m(T) = \sum_{j=0}^m (-1)^{m-j} \binom{m}{j} (T^*)^j T^j.$$

The usual m -isometry condition is $B_m(T) = 0$, with $m \geq 1$. Equivalently, the corresponding alternating binomial sum of $\|T^j v\|^2$ vanishes for every v . The adjoint form follows from $\langle v, (T^*)^j T^j v \rangle = \|T^j v\|^2$ and the polarization identity. The alternative sign convention $(-1)^j$ multiplies the whole identity by $(-1)^m$ and gives the same zero condition.

Let $T(v_0, v_1) = (v_0 + v_1, v_1)$. Its actual matrix is

$$J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Every linear map on this finite-dimensional Hilbert space is bounded. Induction using matrix multiplication gives

$$J^j = \begin{pmatrix} 1 & j \\ 0 & 1 \end{pmatrix}, \quad (J^j)^* J^j = \begin{pmatrix} 1 & j \\ j & j^2 + 1 \end{pmatrix} \quad (j \geq 0).$$

Here conjugate transpose is precisely the Hilbert adjoint in the standard orthonormal basis.

Theorem 1. *T is a 3-isometry, is neither a 1- nor a 2-isometry, and therefore has minimal positive isometry order $3 > \dim_{\mathbb{C}} H = 2$.*

Proof. Write $G_j = (J^j)^* J^j$. Actual matrix computations yield

$$B_1(J) = G_1 - G_0 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \neq 0, \quad B_2(J) = G_2 - 2G_1 + G_0 = \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix} \neq 0.$$

For order three,

$$B_3(J) = G_3 - 3G_2 + 3G_1 - G_0 = 0 :$$

each entry of G_j is a polynomial in j of degree at most two, whose third alternating binomial difference vanishes. These are identities of full matrices, so they hold for all vectors, not just a sample vector. The only positive orders below three are one and two, both excluded above. Thus the minimal positive order is three, contradicting the asserted upper bound by dimension two. $\square$

Formal Hilbert-space correspondence

The Lean project uses the actual Mathlib space `EuclideanSpace  $\mathbb{C}$` (Fin 2), with its Hermitian inner product and Euclidean norm. It does not use the default supremum norm on a function space to define an isometry. `ambient_dimension` proves its actual complex `Module.finrank` is 2. The theorem `jordan_continuous` also verifies the continuity of the actual finite-dimensional operator. The definition `J` is the actual complex matrix displayed above. `jordan_pow` proves its matrix power formula for every natural j . `gram m A` is exactly the full finite sum defining $B_m(A)$, with Mathlib's actual conjugate transpose, matrix powers and multiplication, and natural binomial coefficients cast into $\mathbb{C}$. The theorems `gram_one`, `gram_two` and `gram_three` verify the three full matrix identities. Their nonzero witnesses are respectively entry $(0, 1)$ and entry $(1, 1)$; there are no asserted spectral or norm values. Crucially, the project also transports these matrix computations to actual Hilbert-space linear operators. The definition `representation` is the inverse of Mathlib's star-algebra equivalence `LinearMap.toMatrixOrthonormal`, taken in the actual standard orthonormal basis `EuclideanSpace.basisFun`. This equivalence preserves scalar multiplication, sums, products, powers and adjoints. In the linear endomorphism algebra, `star` is the Hilbert adjoint and multiplication is composition. `operatorGram` is the same binomial identity in that actual endomorphism algebra. `representation_gram` proves that it is the image of the matrix Gram sum, and `gram_operator_iff` proves that one vanishes exactly when the other does. `jordan_is_three` therefore proves that the actual operator is a 3-isometry. `no_smaller_positive_order` quantifies every positive $m < 3$ and excludes that order. The final theorem `dimension_bound_fails` combines genuine 3-isometry, positive-order minimality and the strict actual dimension inequality.

Scope

This is a disproof of the explicitly stated minimal-order dimension-bound conjunct in both languages. The example has positive finite dimension and positive minimal order. No claim about the separate spectrum, unitary criterion, or extension-decomposition clauses is needed to disprove the conjunction. The convention that an m -isometry may also satisfy identities at higher orders causes no ambiguity: we compute its least positive order.

Reproduction

The project pins Lean 4.19.0 and Mathlib commit

```
c44e0c8ee63ca166450922a373c7409c5d26b00b.
```

From `lean/`, run `lake update`, `lake exe cache get`, and `lake build`; then check the entire source with

```
lake env lean -DwarningAsError=true Main.lean.
```

The source prints dependency audits for the main matrix, representation and minimality theorems. No statements are admitted, and only standard logical axioms are used.